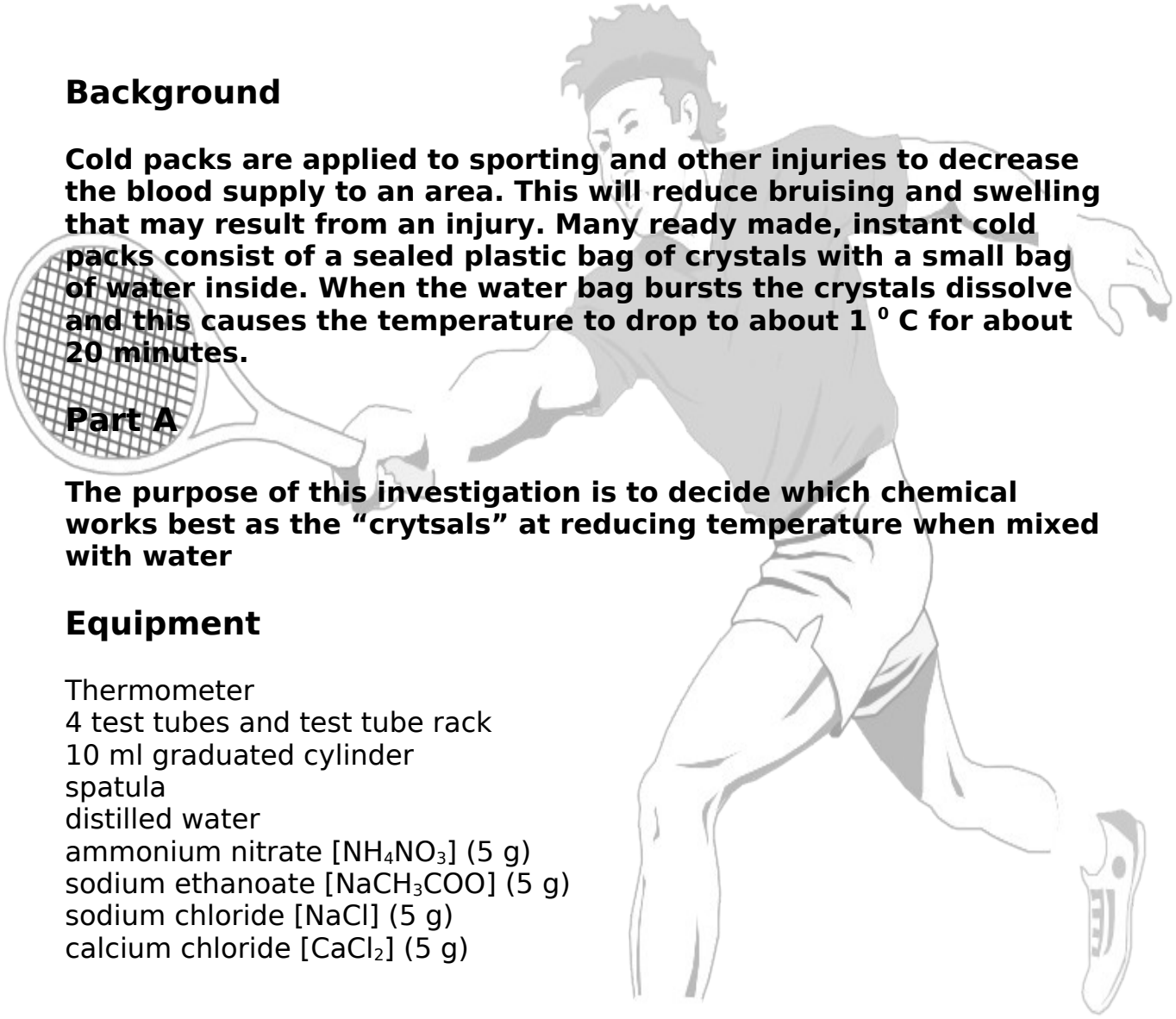
DESIGNING A COLD PACK



Background

Cold packs are applied to sporting and other injuries to decrease the blood supply to an area. This will reduce bruising and swelling that may result from an injury. Many ready made, instant cold packs consist of a sealed plastic bag of crystals with a small bag of water inside. When the water bag bursts the crystals dissolve and this causes the temperature to drop to about 1 ^{0} C for about 20 minutes.

Part A

The purpose of this investigation is to decide which chemical works best as the "crytsals" at reducing temperature when mixed with water

Equipment

Thermometer
4 test tubes and test tube rack
10 ml graduated cylinder
spatula
distilled water
ammonium nitrate [NH_4NO_3] (5 g)
sodium ethanoate [$NaCH_3COO$] (5 g)
sodium chloride [NaCl] (5 g)
calcium chloride [$CaCl_2$] (5 g)

Test to find out which powder would be best suited to making a cold pack. Explain how you tested this and what results you got.



Part B (for the experts)

Extra Equipment

Clean and reuse the 4 test tubes from Part A
A scale or balance measuring in grams

Using only the powder you decided on in Part A, decide what is the optimum mass to use with 30 mls of water. How many moles is this and

what concentration will it give? Once again explain how you tested this and what results you got.